

Dialogue Systems

GUS: Simple Frame-based Dialogue Systems

Dr. Saeedeh Momtazi



دانشگاه صنعتی امیرکبیر
(پلی تکنیک تهران)



آکادمی‌ها



- **Frame-based Dialogue Systems**
- **Control Structure**
- **NLU for Filling Slots in GUS**
- **Other Components of Frame-based Dialogue**



Frame-based Dialogue Agents

Sometimes called "**task-based dialogue agents**"

- Systems that have the goal of helping a user solve a task like
 - ▶ Making a travel reservation
 - ▶ Buying a product

Architecture:

- First proposed of a GUS system in 1977
- A knowledge structure representing user intentions
- One or more **frames** (each consisting of **slots** with **values**)



- A set of **slots**, to be filled with information of a given **type**
- Each associated with a **question** to the user
- Sometimes called a **domain ontology**

Slot	Type	Question
ORIGIN	city	"What city are you leaving from?"
DEST	city	"Where are you going?"
DEP DATE	date	"What day would you like to leave?"
DEP TIME	time	"What time would you like to leave?"
AIRLINE	line	"What is your preferred airline?"



➤ Types in GUS, as in modern frame-based dialogue agents, have hierarchical structure

▶ e.g., the date type in GUS is itself a frame with slots with types like integer or members of sets of weekday names:

DATE

MONTH:NAME

YEAR:INTEGER

DAY:(BOUNDED-INTEGER 1 31)

WEEKDAY:(MEMBER (Sunday Monday Tuesday Wednesday Thursday Friday Saturday))



Two Basic Architectures

➤ The **GUS** architecture

- ▶ Sometimes just called "**frame-based**" architecture
- ▶ Over 40 years old, but still used in most industrial task-based dialogue agents

Bobrow, Daniel G., Ronald M. Kaplan, Martin Kay, Donald A. Norman, Henry Thompson, and Terry Winograd. 1977. "**GUS**, a frame-driven dialog system." Artificial Intelligence 8, 2:155-173.

➤ The **dialogue-state** architecture

- ▶ Extension of GUS
- ▶ More common in research systems
- ▶ Some aspects making their way into industrial systems



Frame-based Dialogue Agents

GUS: Hello. My name is GUS. I can help you plan a simple trip by air. Where do you want to go ?

Client: I want to go to San Diego on May 28

GUS: What time do you want to leave?

Client: I must be in San Diego before 10 am

GUS: Would you like P.S.A. flight 102 that arrives at 9.15 am ?

Client: What is the next flight ?

GUS: Air California flight 310 that leaves at 8.30am and arrives at 10.00am

Client: I'll take the first one

GUS: In what name should I make a reservation, please?

Client: Dan Bobrow

GUS: I have confirmed the following flight: P.S.A. flight 102 on Wednesday May 28 from San Jose to San Diego that leaves at 7.30 am and arrives at 9.15 am.

GUS: What date do you want to return on ?

Client: On Friday in the evening.

GUS: Would you like the flight that leaves at 7.45 pm ?

Client: That's fine.

GUS: I have confirmed the following flight: P.S.A. flight 307 on Friday May 30 from San Diego to San Jose that leaves at 7.45 pm and arrives at 9.30 pm Thank you for calling. Goodbye



- Frame-based Dialogue Systems
- **Control Structure**
- NLU for Filling Slots in GUS
- Other Components of Frame-based Dialogue



Control Structure for GUS Frame Architecture

- System asks questions of user, filling any slots that user specifies
- User might fill many slots at a time:

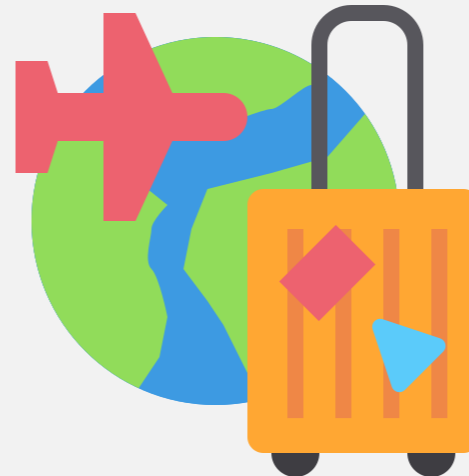
I want a flight from San Francisco to Denver one way leaving after five p.m. on Tuesday.

- When frame is filled, do database query



GUS slots have condition-action rules attached

- Some rules attached to the DESTINATION slot for the plane booking frame
 - ▶ Once the user has specified the destination
 - ▶ Enter that city as the default StayLocation for the hotel booking frame.
 - ▶ Once the user has specified DESTINATION DAY for a short trip
 - ▶ Automatically copy as ARRIVAL DAY.





Multiple Frames

Frames like car or hotel reservations

➤ General route information

- ▶ Which airlines fly from Boston to San Francisco?

➤ Information about airfare practices

- ▶ Do I have to stay a specific number of days to get a decent airfare?

➤ Frame detection:

- ▶ System must detect which slot of which frame user is filling
- ▶ And switch dialogue control to that frame



- Frame-based Dialogue Systems
- Control Structure
- **NLU for Filling Slots in GUS**
- Other Components of Frame-based Dialogue



NLU for Filling Dialog Slots

1- Domain classification

- ▶ Asking weather? Booking a flight? Programming alarm clock?

2- Intent Determination

- ▶ Find a Movie, Show Flight, Remove Calendar Appt

3- Slot Filling

- ▶ Extract the actual slots and fillers



NLU for Filling Dialog Slots

Show me morning flights from Boston to SF on Tuesday.

DOMAIN: AIR-TRAVEL
INTENT: SHOW-FLIGHTS

ORIGIN-CITY: Boston
ORIGIN-DATE: Tuesday
ORIGIN-TIME: morning
DEST-CITY: San Francisco



NLU for Filling Dialog Slots

Wake me tomorrow at six.

DOMAIN: ALARM-CLOCK

INTENT: SET-ALARM

TIME: 2017-07-01 0600



How to fill slots? (1) Rule-based Slot Filling

➤ Write regular expressions or grammar rules

Wake me (up) | set (the|an) alarm | get me up

➤ Do text normalization



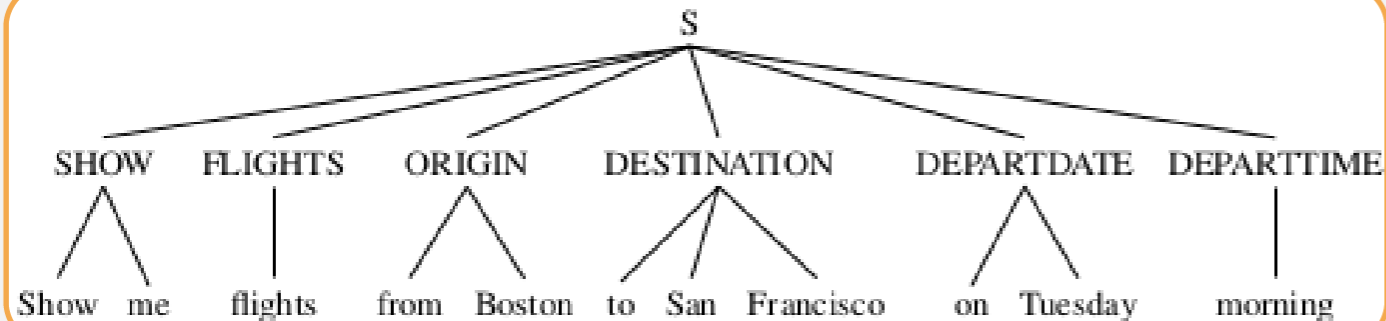
Rule-based Slot Filling

- Rule-based research systems like the Phoenix system (Ward and Issar, 1994) consist of large hand-designed semantic grammars with thousands of rules
- A semantic grammar is a context-free grammar in which the left-hand side of each rule corresponds to the semantic entities being expressed (i.e., the slot names)
- Semantic grammars can be parsed by any CFG parsing algorithm, resulting in a hierarchical labeling of the input string with semantic node labels



Rule-based Slot Filling

SHOW	→ show me i want can i see ...
DEPART_TIME_RANGE	→ (after around before) HOUR morning afternoon evening
HOUR	→ one two three four... twelve (AMPM)
FLIGHTS	→ (a) flight flights
AMPM	→ am pm
ORIGIN	→ from CITY
DESTINATION	→ to CITY
CITY	→ Boston San Francisco Denver Washington





- Many industrial dialogue systems employ the GUS architecture
- But use supervised machine learning for slot-filling instead of these kinds of rules
- We will discuss later



- Frame-based Dialogue Systems
- Control Structure
- NLU for Filling Slots in GUS
- **Other Components of Frame-based Dialogue**



Generating Responses: Template-based Generation

➤ A template is a pre-built response string

▶ Templates can be fixed:

"Hello, how can I help you?"

▶ Or have variables:

"What time do you want to leave CITY-ORIG?"

"Will you return to CITY-ORIG from CITY-DEST?"



- A simple frame-based architecture
- Like many rule-based approaches
- Positives:
 - ▶ High precision
 - ▶ Can provide coverage if the domain is narrow
- Negatives:
 - ▶ Can be expensive and slow to create rules
 - ▶ Can suffer from recall problems



Further Reading

➤ Speech and Language Processing (3rd ed. draft)

▶ Chapter 24